

Om Pranab Mohanty

Odisha, India | +91 9938827986 | ompranabmohanty@gmail.com | GitHub | LinkedIn | Portfolio

Undergraduate in Mathematical Sciences & Computing

Focus: Computer Architecture, Operating Systems, Performance Engineering, Mathematical Modeling

EDUCATION

Sri Sathya Sai Institute of Higher Learning — Puttaparthi, India Jul 2023 – Present

B.S. (Hons. with Research) in Mathematical Sciences and Computing

Specialization: Artificial Intelligence

CGPA: 8.4 / 10 (O Grade) Rank 1 in cohort

Sri Sathya Sai Higher Secondary School — Puttaparthi, India Jul 2021 – Apr 2023

Intermediate Education (Class XII)

Subjects: Mathematics, Physics, Chemistry, Computer Science, English

Score: 85 / 100

WORK EXPERIENCE

Research Intern — Manipal Institute of Virology Mar 2026 – Present

- Worked on VAE-based modeling of long-term genomic data for dengue serotype analysis
- Focused on large-scale data representation and computational constraints

PROJECTS

LowBit: Lightweight Benchmarking and Instrumentation Toolkit [Page] Jul 2026 – Present

- Developed an open-source suite of lightweight, deterministic microbenchmarks for computer architecture and machine learning performance evaluation.
- Released **LB-ML-001**, implementing the Transformer scaled dot-product attention kernel with reproducible workloads, reference validation, benchmark metadata, and CMake-based cross-platform builds.
- Designed the benchmark methodology, documentation, validation framework, and microarchitecture analysis pipeline to support future benchmark development within the LowBit ecosystem.

OS Synchronization as Symmetric Restriction in S_n [Code] Feb 2026 – Present

- Modeled OS synchronization (mutex, scheduling, deadlock) using group-theoretic structures in S_n
- Established safety–fairness trade-offs and introduced a metric for scheduling transitions
- Implemented validation framework in Python for small-scale systems

Network Ingress Subsystem in C [Code] Mar 2026 – Apr 2026

- Built a deterministic packet processing system using fixed-size buffers and protocol parsers
- Ensured predictable execution by eliminating dynamic memory allocation
- Validated behavior under constrained system conditions

SKILLS

Systems & Architecture — Computer Architecture, Operating Systems, Computer Networks,

Programming — C/C++, Python, SQL, Bash

Mathematics — Discrete Mathematics, Linear Algebra, Optimization, Probability

Tools — Git, Linux

COURSEWORK

Systems — Operating Systems, Computer Architecture, Computer Networks

AI — Data Mining and ML, Deep Learning, Natural Language Processing

Mathematics — Discrete Mathematics, Linear Algebra, Optimization, Abstract Algebra, Probability

Core CS — Algorithms

EXTRACURRICULAR ACTIVITIES

Secretary — Speech Craft Society, SSSIHL Feb 2025 – July 2026

- Led organization of student sessions on public speaking and structured communication